GE 213

Introduction to Engineering Design II

2010/2011

Introduction to Microsoft Excel

Part II

Excel Logical Functions

Table of Content

Introduction	**1**
AND Logical Function.	**1**
2.1. Example 1.	**2**
OR Logical Function.	**3**
3.1. Example 2.	**4**
NOT Logical Function.	**5**
4.1. Example 3.	**6**
IF Logical Function.	**7**
5.1. Example 4.	**6**
Example Problem.	**8**
References	**9**

1. Introduction

The ***aim*** of this document is to introduce guidelines of using logical functions in the Microsoft Excel environment. Students *should* follow the instructions to create the conditional formulas and *must* practice an understanding of formulating by logics; therefore, it is very important to understand the logical function before implementing it in the spreadsheet.

Each section in this document start with focusing on the ***logical function*** *meaning* and *format*, this followed by a systematic instructions to practice *implementing* this logical function at the Excel environment.

Before using this document, you have to check some options in your Microsoft Excel. First, check that the calculation is ***Manual*** by choosing Tools → Options → Calculation and then select Manual. Second, at the same calculation window select ***Iteration***, set the Maximum Iteration equal 1, and set the Maximum Change equal 1.

2. AND Logical Function /1/.

AND returns **TRUE** if ***all*** its arguments are TRUE; returns **FALSE** if ***one or more*** argument is FALSE. Regarding figure (1), if the two conditions are satisfied then the output will be true, and, the output will be false if one of the inputs is false.

AND (**logical1**,logical2,...),where Logical1, logical2,...are 1 to 30 conditions you want to test that can be either TRUE or FALSE.

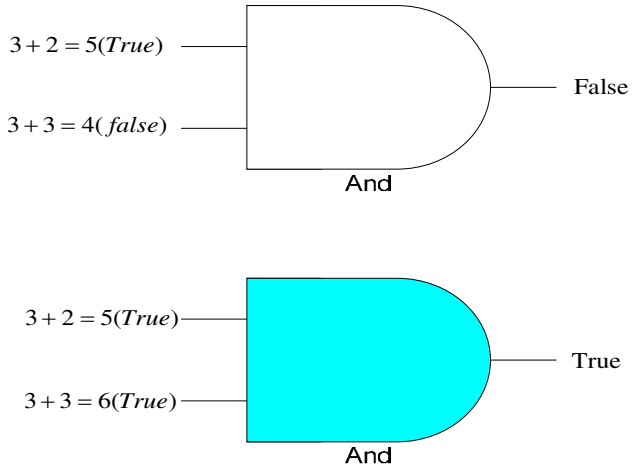


Figure (1) AND logical function

2.1. Example 1.

Open a new worksheet and enter A1=7, A2=8, A3=9 and A4=10, see figure (1), now at the cell A7 write the following logic A7=AND(A2>A1;A3>A2) then press Enter. The word True will appear, see figure (3). The word "True" will appear because the two conditions A2>A1 and A3>A2 are true. Now, let A8 =AND(A2<A1;A4>A3) and press Enter. The word "False" will appear because at least one of the conditions is not true, you can see that A2 is not less than A1.

The "And" logic function inputs could be equations, see cell A9. Also "And" logic function may get more than two inputs, see cell A11.

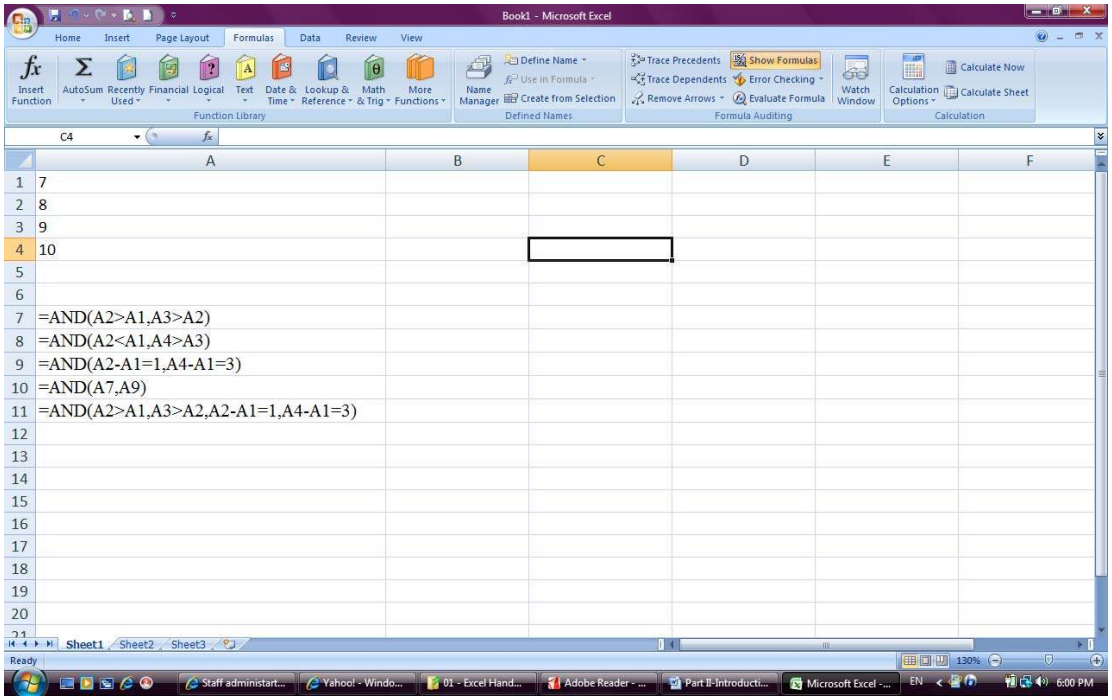


Figure (2) Spreadsheet Example for the "And" logic (Formula option)

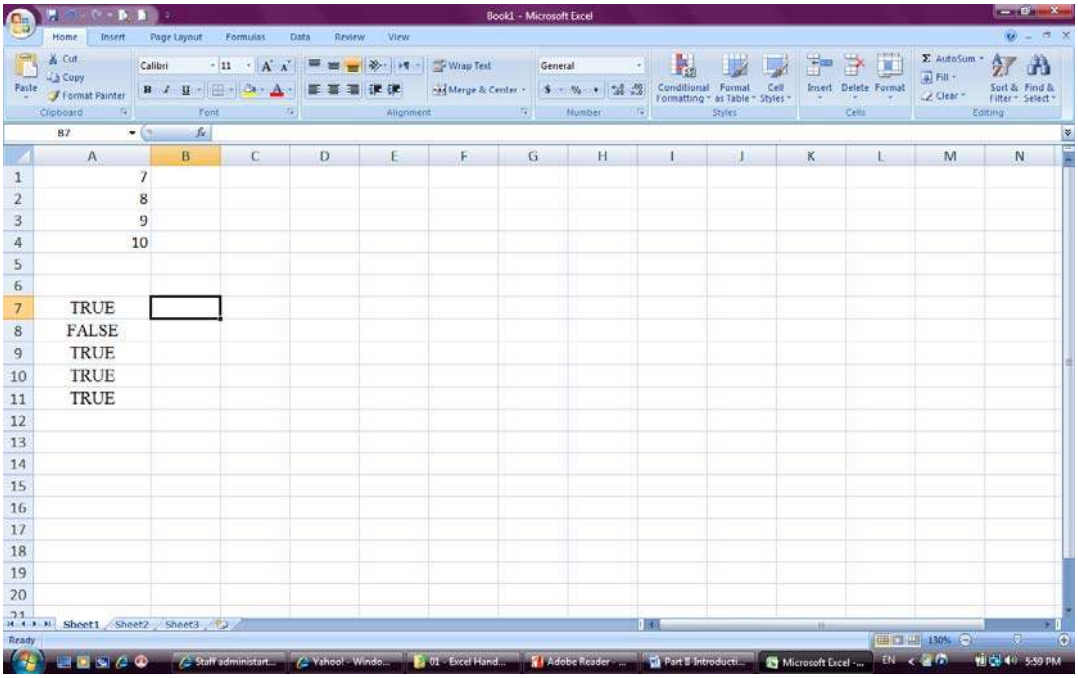


Figure (3) Spreadsheet example for the "And" logic

3. OR Logical Function /1/.

OR returns **TRUE** if ***any*** argument is TRUE; returns **FALSE** if ***all*** arguments are FALSE. Regarding figure (4), one of the conditions is TRUE, so, the output will be TRUE.

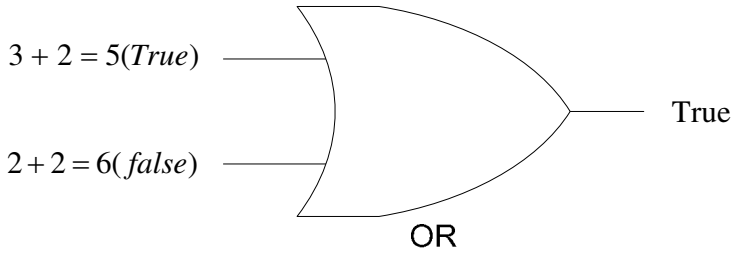


Figure (4) OR logical function

OR (**logical1**,logical2,...),where Logical1, logical2,...are 1 to 30 conditions you want to test that can be either TRUE or FALSE.

3.1. Example 2.

Create two random numbers between 1 and 10 and check if either the first number is greater than 5 or the second is greater than 6

As shown in figure (5), let cell A1=**RAND**()*10 this will create a random number between 0 and 1 multiplied by 10. At cell A2 type "=RAND()*10", this will create another random number between 0 and 10. Now at cell A5 type "=OR(A1>5;A2>6)".

Now each time you will press **F9** a couple random numbers will be generated at cells A1 and A2. At each time you press **F9** you will see cell A5 either **TRUE** or **FALSE**. It will be ***TRUE*** when ***at least one*** of the logics A1>5, A2>6 **occurs**. Figure (6) shows a few sample runs.

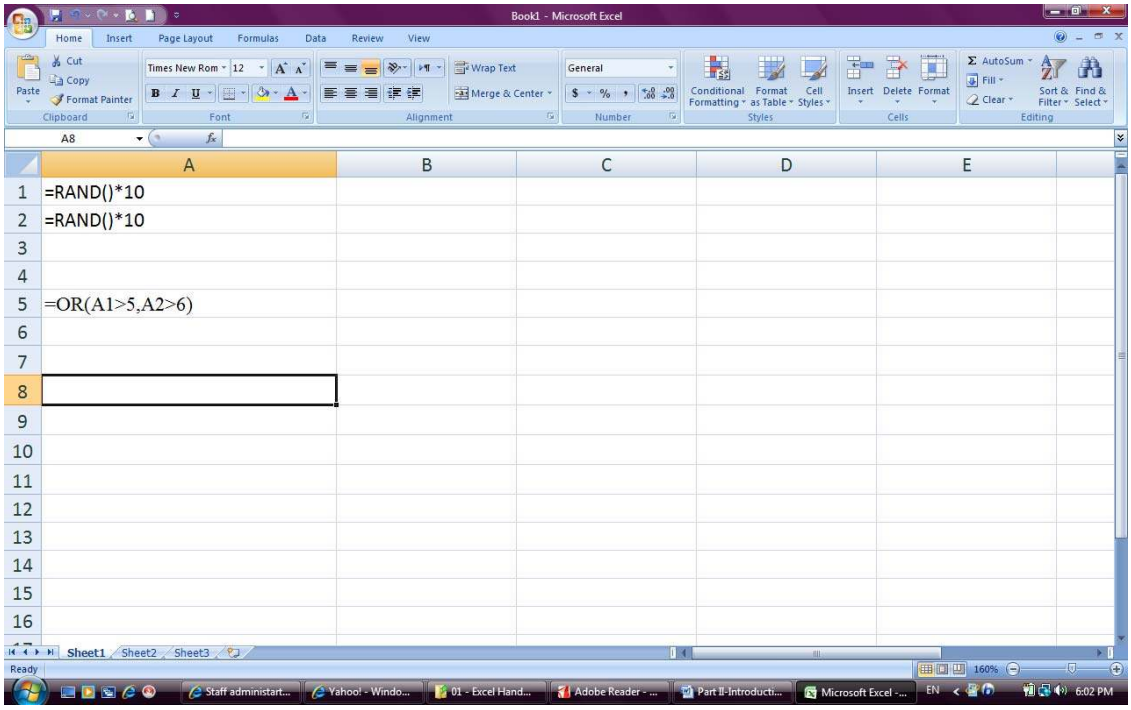


Figure (5) Spreadsheet of example 2 (Formula option)

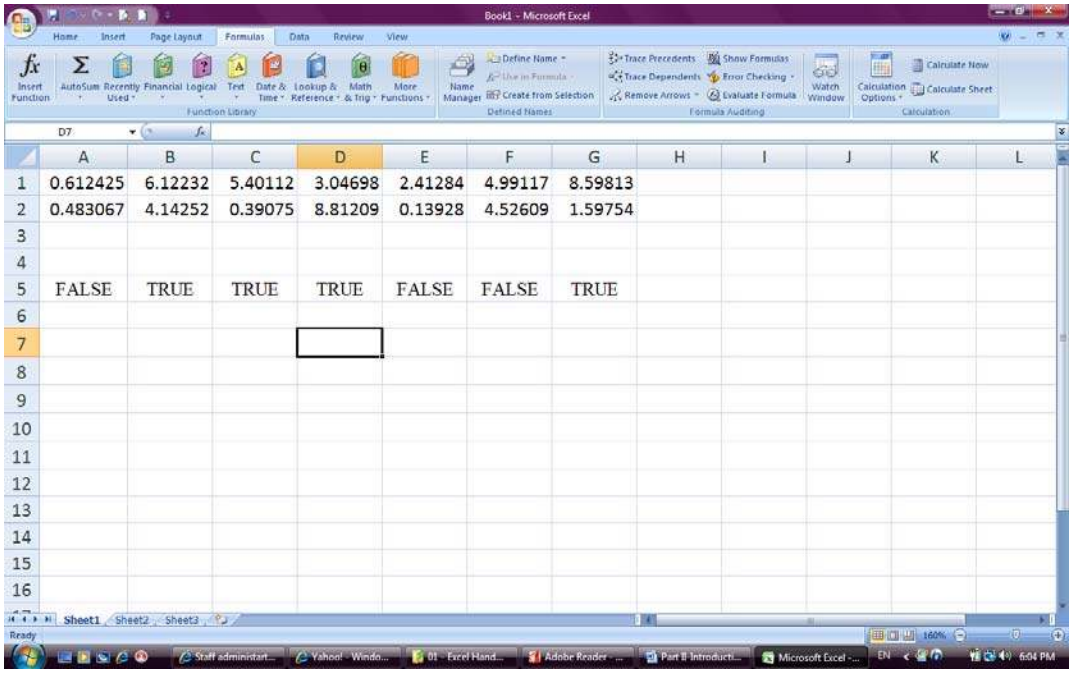


Figure (6) Example 2 multi-runs

4. NOT Logical Function /1/.

NOT reverses the value of its argument. Use NOT when you want to make sure a value is not equal to one particular value. If logical is FALSE, NOT returns TRUE; if logical is TRUE, NOT returns FALSE.

4.1. Example 3.

Continue at Example 2 type at cell A8 " =NOT(A5)". This means that if cell A5 is TURE, cell A8 will be False. See figure (7).

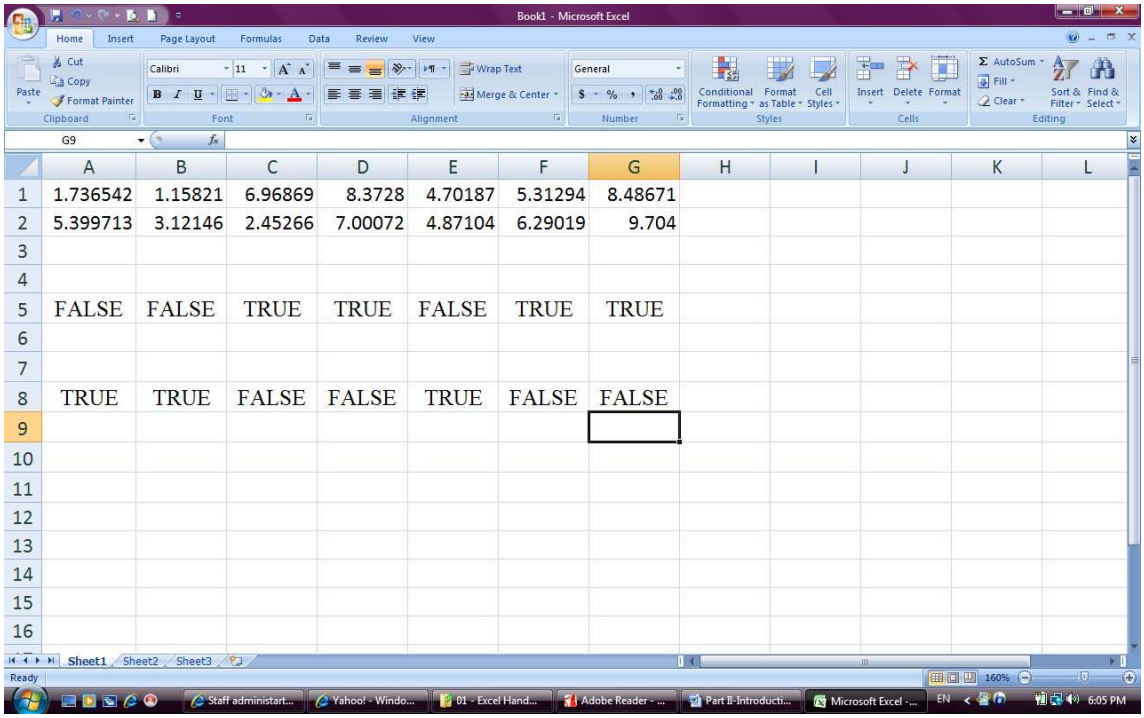


Figure (7) Example 3, NOT logical function

5. IF Logical Function /1/.

IF returns one value if a condition you specify evaluates to TRUE and another value if it evaluates to FALSE. Use IF to conduct conditional tests on values and formulas.

IF (logical test ; Value if true ; Value if false)

→ ***Logical test*** is any value or expression that can be evaluated to TRUE or FALSE. For example, A10=100 is a logical expression; if the value in cell A10 is equal to 100, the expression evaluates to TRUE. Otherwise, the expression evaluates to FALSE.

→ ***Value_if_true*** is the value that is returned if logical_test is TRUE.

→ **Value_if_false** is the value that is returned if logical_test is FALSE.

The *Value if true* and *Value if false* could be a ***text***, ***value*** and ***logical function***. The following example will indicate the three cases.

5.1. Example 4.

Create two integer random numbers between 1 and 3 and write a logical function that give a sign if both random numbers are equal 2

As shown in figure (8), at cell A1 type"=ROUNDUP(RAND()*3)" This will create an integer number between 1 and 3. Write the same at cell A2, this will create another random number between 1 and 3. Now, each time you press F9 the numbers will change. At cell A4 type "=IF(A1=A2;"Equality";"Inequality")" this logic means that if A1=A2 the word **Equality** will appear and otherwise the word **Inequality** will appear. Now at cell A5 type "=IF(AND(A4="EQUALITY";A1=2);1;0)" this logic means that if the word **Equality *and* A1=2** both occur, number 1 will appear at cell A5 otherwise number 0 will appear.

One logic can do the same task, if we type at cell A6 type "=IF(A1=A2;***(IF(A1=2;1;0))***;0)". This means that if A1=A2 another logic will be tested " IF A1=2 or not, and, if this test is succeed number 1 will appear otherwise number 0 will appear and if A1≠ A2 number 0 will appear at cell A6.

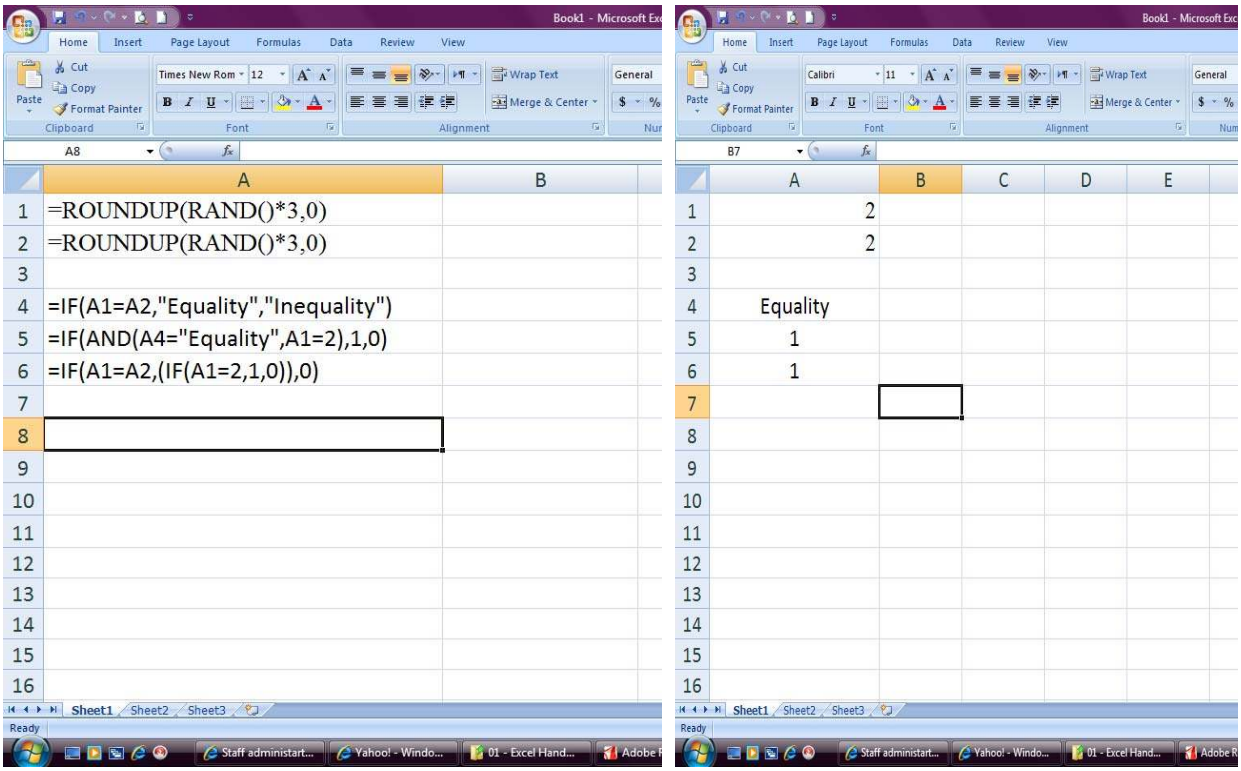


Figure (8) Example 4, IF logical function

6. Example problem.

Determination of the value of π using an iterative Excel approach:

If a circle of radius R is inscribed inside a square with side length 2R, as shown in figure (9) Then the area of the circle will be $\pi*R^2$ and the area of the square will be $(2R)^2$. So the ratio of the area of the circle to the area of the square will be π/4.

This means that, if you pick N points at random inside the square, approximately N*π/4 of those points should fall inside the circle.

Construct a program that, picks points at random inside the square. It then checks to see if the point is inside the circle (it knows it's inside the circle if $x^2 + y^2 < R^2$, where x and y are the coordinates of the point and R is the radius of the circle). The program keeps track of how many points it's picked so far (N) and how many of those points fell inside the circle (M). π is then approximated as shown in equation (1).

$$\pi = \frac{4M}{N} \qquad (1)$$

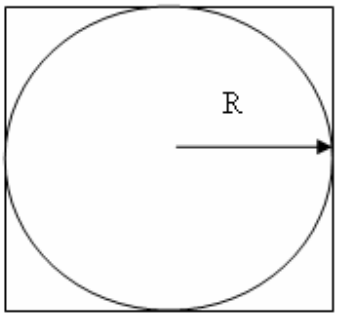


Figure (9) A circle of radius R is inscribed inside a square with side length 2R

Model:

At cell B1 type the problem title “Excel Model To Calculate the Value of Pi Using an Iterative Approach” as shown in figure (10). At cell A4 type “R =” then type “L =” at cell A5 .Then enter the values R=”1” and L =”2” at cell B4 and B5 respectively.

Now we want to select a point inside the square, also, we need a ***counter*** that counts the number of selected points together with a ***reset*** to this counter. At cell A7 type "Reset =" and type 1 at cell B7 after renaming it as **Reset**. Now type "Iteration (N)" at cell A8 and rename cell B8 as **Iterations**. Type inside the cell B8 "=IF(Reset=0;0;itr+1)". Now each time you will press F9 the number in B9 will increase by 1, this will count the number of times you select a random point inside the square or count N in equation (1). If the Reset="0" the value at Iterations box will reset to zero also.

We want to select a point inside the square so we want to create an x and y values for the point .Assuming that the center of the coordinates located at the center of the square. For the point (x,y) x and y range from 1 to -1, so, if we want to select a random x for the point we must select a random sign and then a random number between 0 and 1. The random sign should be either + or – , in other words, value of x multiplied by 1 or -1.

At cell A12 type "X - Sign" and at cell B12 type "=Reset*RAND()". Now you created a random number between 0 and 1. So if the random number between 0 and 0.5 the sign will be - and if the random between 0.5 and 1 the sign will be +. At cell C11 type "=Reset*IF(B12<=0.5;-1; 1)". Do the same for the Y sign at cells A13, B13, and C13 as shown in figure (10).

Now we want to select a random x and y values with a random sign. Type "=Reset*RAND()*C12*B4" at cell B17 to chose a random x value multiplied by a random sign 1 or -1 from cell C12 and multiplied by B4. X-value will be always inside the square. Do the same for Y at cell B18 as shown in figure (10).

We want now to check if the point (x,y) inside the circle or not. At cell B22 type" = Reset * IF(($B$17^2+$B$18^2)<=$B$4^2, 1,0)" this will evaluate x^2+y^2. Then if $x^2+y^2<1$ we will count 1 or count a point inside the circle. These counted points could be summed at cell B23 by typing = IF(Reset=0,0,M+B22)" to obtain the value of M .

Now we have the value of N from Cell B8 and M from cell E20. To count π by Appling equation (1) type "= IF(Reset=0,0, 4*M/itr)" at cell H12.

Now but Reset=0 at cell B7 then press F9. Then enter 1 at the same cell and start the calculations by pressing F9. Each time you press F9 the Iterations will increase by 1 and a random point of a random x and y will be created and checked to see if it is inside the circle or not.

The value of π will appear at cell H12 and each time we will press F9 it will become closer to the exact value of π=3.1417…. as shown in figure (11).

7. References.

1. Microsoft ® Office Excel 2007, Microsoft Excel Help, Copyright © 1985-2007 Microsoft Corporation.

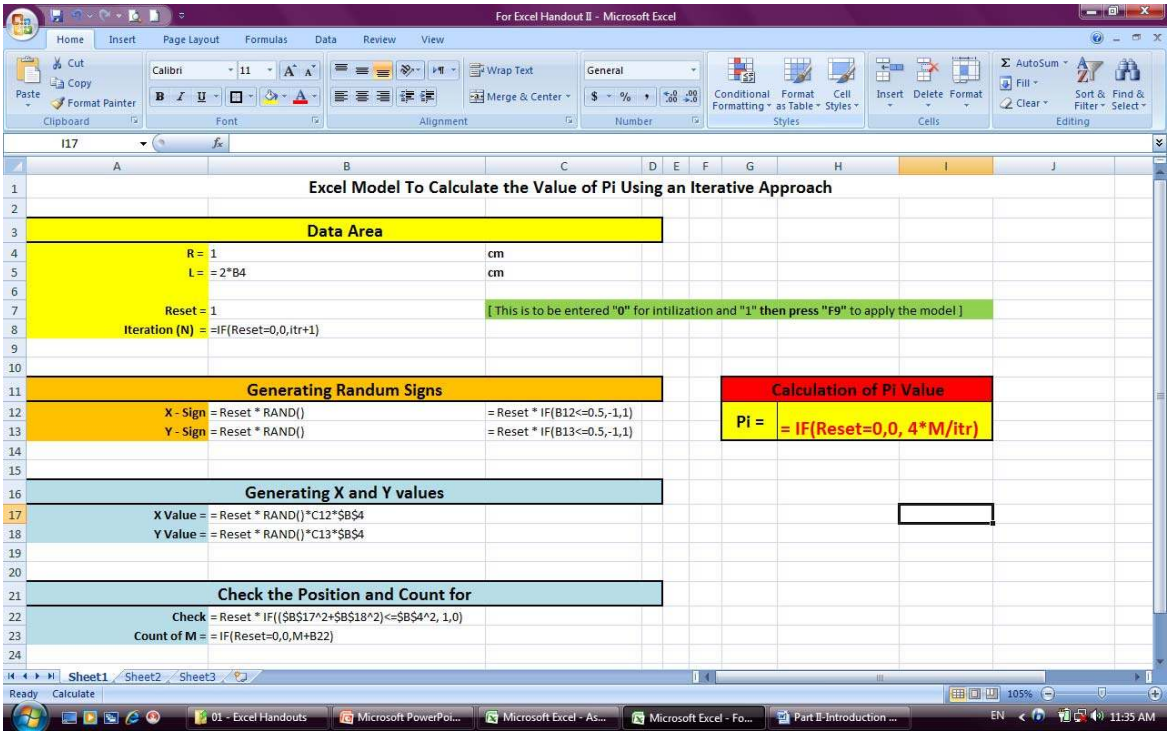


Figure (10) Problem 1 Formula sheet

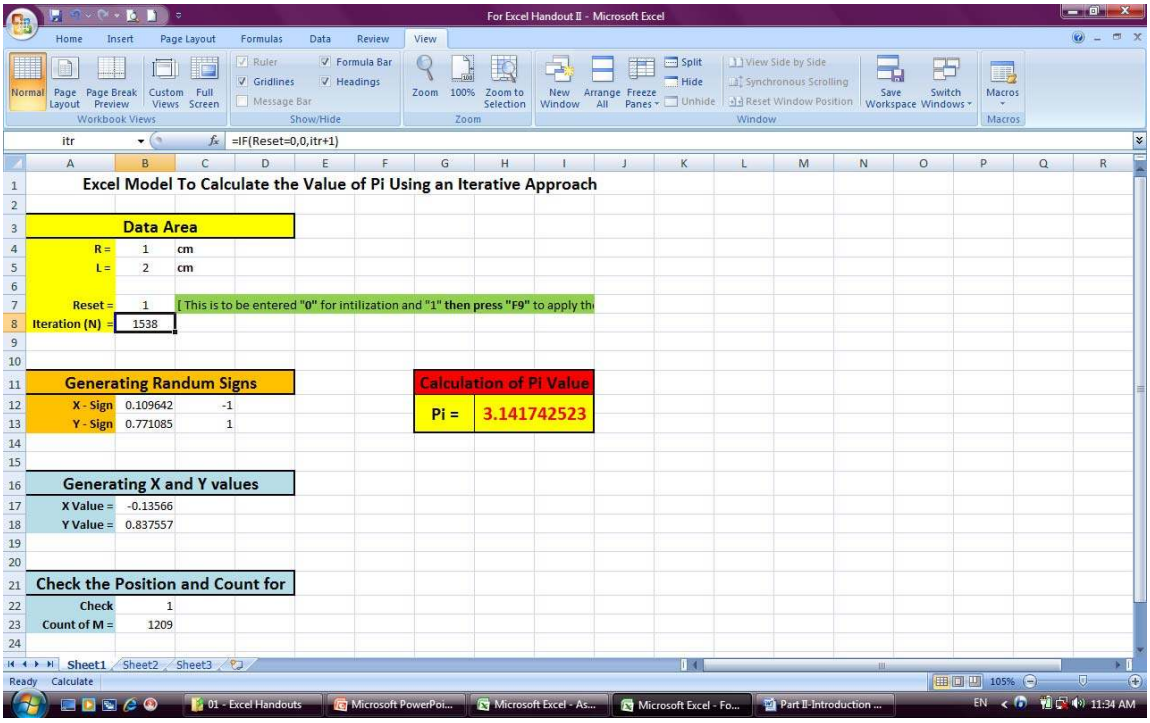


Figure (11) Values sheet